DBMS LAB — QUESTION 7

AIM

Create and query views and materialized views using appropriate SQL commands.

DESCRIPTION

A VIEW is a virtual table based on a SELECT query. It normally stores the query definition rather than a separate copy of the result.

A MATERIALIZED VIEW stores the result of its query. It is useful for repeatedly accessed or expensive summary queries and can be refreshed when required.

The program creates an IT employee view and a department-wise salary materialized view, then queries both.

CODE

```
CREATE TABLE Employee (
    EmpID NUMBER PRIMARY KEY,
    EmpName VARCHAR2(50),
    Department VARCHAR2(30),
    Salary NUMBER(10,2),
    City VARCHAR2(30)
);

INSERT INTO Employee VALUES (201,'John','HR',50000,'Hyderabad');
INSERT INTO Employee VALUES (202,'Alice','IT',60000,'Chennai');
INSERT INTO Employee VALUES (203,'Bob','Finance',55000,'Bangalore');
INSERT INTO Employee VALUES (204,'Emma','IT',65000,'Pune');
INSERT INTO Employee VALUES (205,'David','Marketing',48000,'Delhi');
INSERT INTO Employee VALUES (206,'Sophia','IT',70000,'Chennai');
COMMIT;

CREATE VIEW IT_Employees AS
SELECT EmpID, EmpName, Salary, City
FROM Employee
WHERE Department = 'IT';

SELECT * FROM IT_Employees;

CREATE MATERIALIZED VIEW MV_Department_Salary AS
SELECT Department, COUNT(*) AS Employee_Count,
       SUM(Salary) AS Total_Salary,
       AVG(Salary) AS Average_Salary
FROM Employee
GROUP BY Department;

SELECT * FROM MV_Department_Salary;

EXEC DBMS_MVIEW.REFRESH('MV_Department_Salary');

SELECT * FROM MV_Department_Salary;
```

EXPECTED OUTPUT

EMPID	EMPNAME	SALARY	CITY
202	Alice	60000	Chennai
204	Emma	65000	Pune
206	Sophia	70000	Chennai

Normal View output:

DEPARTMENT	COUNT	SUM	AVG
HR	1	50000	50000
IT	3	195000	65000
Finance	1	55000	55000
Marketing	1	48000	48000

Materialized View output:

IMPORTANT NOTES

DBMS_MVIEW.REFRESH is the Oracle refresh command used here.

Dropping a view does not delete the underlying Employee table.